

ERGT: Attention-Free Native Geodesic Transport for Controlled Long-Horizon Relational Reasoning

Jalal Jafari

Ph.D. in Photonics; Board Member, NovoXpert AI Group
Artificial Intelligence Research and Development
j.jafari2009@gmail.com

Nasibeh Aramideh

Ph.D. in Mathematics; Research Member, NovoXpert AI Group
Artificial Intelligence Research and Development
aramidehnasibeh@gmail.com

Abstract

In this paper, we introduce Executable Relational Geometry and Transport (ERGT), an attention-free architecture for learning relational structure directly from raw input. ERGT constructs a persistent information field (Ψ), a stable representation from which pairwise observables induce multiple directed world geometries. GeoTransport propagates typed states along admissible paths within these geometries. A geodesic solver identifies the minimum-cost admissible path, while boundary measurement accepts a candidate only if the transported state satisfies the terminal constraints. Unlike conventional approaches, ERGT requires neither Transformer architectures nor externally supplied graphs, target compilers, or generic execution engines. Instead, both the representational structure and the inference process emerge directly from the input data.

Across four independent runs, ERGT converged consistently and satisfied all predefined stability criteria. Beyond its training horizon, the model achieved a mean accuracy of 0.955 at 32 hops, compared with 0.500 for a stronger Transformer baseline evaluated on the same raw input with a closely matched parameter count. Ablation experiments showed substantial performance degradation when geodesic action, propagation admissibility, typed transport, geometric memory, or boundary measurement was removed. Likewise, randomizing the learned geometry or restricting inference to a single world failed to recover the original performance. These findings support the claim that a persistent information field (Ψ) constructed directly from raw input can induce an executable relational geometry, providing a measurable advantage for controlled long-horizon relational reasoning.

Keywords: relational reasoning, representation learning, information geometry, attention-free architecture, geodesic transport, long-horizon generalization, causal intervention

1 Introduction

Transformer architectures have demonstrated the effectiveness of content-dependent token interactions for representation learning, with these interactions realized through self-attention [17]. However, long-horizon relational reasoning imposes additional structural requirements. Entities must remain distinguishable, directed and typed relations must compose in a well-defined order, and intermediate state must be preserved across repeated propagation steps. Furthermore, the model should abstain from producing an answer when no valid support path exists. Although such behaviors may emerge

through training, conventional sequence architectures do not encode admissibility, conservation, or principled rejection as intrinsic computational primitives.

Executable Relational Geometry and Transport (ERGT) is built around this distinction. Its central premise is that the arrangement of information in an input can generate the geometry on which reasoning is performed. Rather than starting from attention scores or a task-supplied graph, the model constructs a persistent information field Ψ directly from raw tokens. This field maintains entity identity and relational evidence within a common source state, providing the substrate from which all subsequent geometry is derived.

Pairwise observables of Ψ induce multiple directed world geometries, or *worlds*, each capturing a distinct relational tendency. The term is purely operational: a world contributes support, direction, and traversal cost, rather than representing a literal physical universe. GeoTransport propagates typed states along admissible paths while preserving entity identity, operator state, payload, and memory. A geodesic solver identifies the minimum-cost admissible path, and boundary measurement accepts a candidate only if the transported state satisfies the terminal constraints. The resulting computational pipeline is

$$\begin{aligned} \text{raw tokens} &\longrightarrow \Psi \longrightarrow \text{world geometries} \\ &\longrightarrow \text{GeoTransport} \longrightarrow \text{geodesic solution} \longrightarrow \text{boundary decision.} \end{aligned}$$

No graph or geometry is provided at the input boundary, and the native inference path does not delegate decision making to a generic compiler or graph execution engine. Instead, relational support, path costs, transported state, and boundary acceptance all emerge from the persistent information field constructed from the same raw input sequence.

The design addresses a characteristic failure mode of long-horizon relational reasoning. As typed relational dependencies accumulate, repeated global reorganization can weaken the association between entity identity, computational state, and propagation history. Rather than repeatedly recomputing a global representation, ERGT composes reasoning through local transport governed by shared admissibility and conservation constraints. Its inductive bias is not merely a larger context window, but the preservation of executable relational paths as reasoning depth increases.

The experimental evaluation follows the same logic. It first evaluates whether the complete inference pipeline, from persistent information field construction to boundary measurement, converges and satisfies the predefined stability criteria. Frozen checkpoints are then evaluated on newly generated relational chains extending well beyond the training horizon. Finally, targeted ablations, geometry corruption, and single-world inference evaluate whether the observed performance depends on the proposed computational mechanism.

The primary baseline is a strengthened Transformer baseline trained on the same raw input sequences and target families. Staged training together with auxiliary supervision maximizes its opportunity to learn the task, while inference remains entirely Transformer-based. The baseline receives no ERGT-derived geometry, transport state, compiler, execution engine, or identity side channel. This preserves a controlled comparison between two complete inference strategies without transferring components of the proposed architecture to the baseline.

The contributions of this work are:

1. introducing a persistent information field (Ψ) constructed directly from raw tokens, from which pairwise observables induce relational geometry without relying on externally supplied structure;
2. developing an end-to-end, attention-free inference pipeline combining directed world geometries, finite-speed admissibility, typed GeoTransport, geometric memory, min-plus geodesic closure, boundary measurement, and fail-closed inference;
3. establishing a controlled multi-seed evaluation protocol that separates learnability from long-horizon extrapolation while comparing ERGT with a strengthened Transformer baseline under the same raw-input boundary; and

4. providing frozen causal interventions together with a self-contained reproduction package, enabling the independent evaluation of the roles of geometry, transport, memory, boundary conditions, and multiworld composition.

The scope of our claim is deliberately bounded. ERGT is not proposed as a universal replacement for Transformer architectures, a general-purpose language model, or a lower-cost sequence architecture. Instead, this work asks whether geometry induced directly from raw tokens can serve as an executable, mechanistically testable inductive bias for controlled long-horizon relational reasoning. Section 5 reports the corresponding empirical results, uncertainty estimates, and effect sizes.

2 Positioning and Related Work

2.1 Attention and attention-free sequence backbones

Removing attention is not, by itself, a new research direction. Self-attention computes content-dependent interactions across sequence positions [17], whereas S4 models long-range dependencies through structured state-space dynamics [12]. Mamba makes those dynamics input-selective [11], and Hyena employs implicit long convolutions with data-controlled gating [15]. Taken together, these architectures demonstrate that sequence modeling can be performed effectively without explicit self-attention.

Their primary computational object remains the sequence operator: recurrence, state-space dynamics, or convolution determines how information is propagated across sequence positions. ERGT addresses a different computational object. It first constructs a token-indexed persistent information field Ψ , from which pairwise observables induce directed world geometries. GeoTransport then propagates typed states through these geometries toward a boundary decision. The contribution is therefore not attention removal alone, nor a claim about language-modeling quality or linear-time efficiency, but a raw-input relational geometry whose constraints participate directly in the inference process.

2.2 Graph reasoning, path methods, and latent structure

Graph neural networks introduce relational inductive bias through message passing [2, 9], while geometric deep learning provides a broader framework for learning and computation on non-Euclidean domains [4]. In most existing approaches, however, the underlying relational structure is provided as part of the input. NBFNet is particularly close in spirit because it represents a node pair through paths and learns a generalized Bellman–Ford computation for graph reasoning [19]. Nevertheless, its paths are still defined over an explicitly supplied graph.

Other approaches learn latent relational structure from observations. Neural Relational Inference (NRI), for example, infers an interaction graph while predicting the dynamics of interacting systems [13]. The objective of ERGT is more specific than simply learning a graph. It requires relational structure to emerge directly from raw tokens and become immediately executable through directed world geometries, geodesic action, typed transport, admissibility, memory, and boundary measurement. In this framework, the persistent information field Ψ is not an additional embedding layer but the canonical computational source from which entity identity, pairwise observables, and every world geometry are derived. Introducing externally supplied edges, entity identifiers, or target graphs would violate that computational contract.

2.3 Neural algorithmic reasoning and size generalization

The CLRS benchmark established size generalization as a central evaluation criterion for neural algorithmic reasoning [18]. Hint-ReLIC further demonstrates the importance of intermediate compu-

tation by regularizing internal states to remain invariant under interventions that preserve the next algorithmic step [3]. Together, these works caution against treating final accuracy alone as sufficient evidence of a reusable computational mechanism.

ERGTadopts a similar evaluation principle without relying on supplied execution traces or intermediate hints. Instead, it learns its geometric dynamics directly from the input sequence and evaluates whether the same local transport and admissibility rules continue to compose at greater reasoning depths. After checkpoint selection, individual geometric components are removed, shuffled, or restricted while the model parameters remain fixed. The resulting performance changes evaluate whether each component is required by the learned inference mechanism, rather than merely benefiting from explicit supervision during training.

2.4 Physics-informed inductive bias and information geometry

Physics-informed models illustrate the value of encoding desired structural properties as architectural constraints rather than expecting them to emerge solely from statistical correlations. Hamiltonian Neural Networks encode conservation laws in learned dynamics [10], while explicit Hamiltonian and Lagrangian constraints extend the same principle [7]. Geometric deep learning exploits symmetry and geometry to restrict the hypothesis space [4], while information geometry provides a mathematical framework for metrics and dual structure on statistical manifolds [1]. ERGTadopts these ideas as design analogies rather than as a direct statistical-manifold construction.

ERGTborrows the underlying design logic rather than a literal physical interpretation. Its terminology has explicit computational meaning: Ψ stores persistent token information, world geometries define directed relational structure, geodesic action determines traversal cost, the propagation cone enforces admissibility, and GeoTransport moves typed states. A geodesic corresponds to the least-cost admissible route, conservation preserves payload identity, and boundary measurement determines acceptance.

Spectral graph quantities [5] and a Forman-style curvature proxy [8] are intentionally kept outside the native inference pipeline. They characterize how the induced geometry evolves with reasoning horizon and motivate future work on curvature-guided routing and regularization [16]. They are not used to explain or attribute the empirical performance reported in this work.

2.5 Positioning of the proposed contribution

Table 1 summarizes the relationship between ERGTand neighboring research directions. The distinguishing feature is not any individual familiar component, but the computational object that the model constructs and executes.

No individual component constitutes the novelty claim. Shortest-path algorithms, graph learning, conservation principles, attention-free propagation, and boundary conditions all have established precedents. Instead, the contribution lies in their integration under a single auditable computational contract. Raw tokens first construct a persistent information field Ψ ; pairwise observables of that field induce multiple directed world geometries; shared admissibility and conservation constraints govern typed transport; an admissible geodesic is composed; and boundary measurement alone determines whether an answer is accepted. This geometry-first contract, rather than general superiority on general-purpose sequence modeling, defines both the proposed contribution and the empirical burden of the paper.

Table 1: Position of ERGT relative to neighboring research families.

Family	Primary computational object	Relation to ERGT
Transformer	Content-dependent token interactions.	Direct token mixing rather than executable relational geometry and geodesic reasoning.
S4, Mamba, Hyena	State-space or long-convolution sequence modeling.	Replace the sequence operator rather than constructing executable geometry.
GNNs, NBFNet	Message passing on graph-structured data.	Typically assume an explicit graph; ERGT induces geometry directly from raw tokens.
Latent graph learning	Learning hidden relational structure.	Further equips the induced structure with admissibility, geodesic action, typed transport, and boundary verification.
CLRS, Hint-ReLIC	Learning algorithm execution.	Native geometric execution without external execution traces at inference.
Physics-informed learning	Learning under physical constraints.	Uses physics-inspired concepts as operational computational constraints.
ERGT	Persistent information field and executable induced geometry.	Integrates structure induction, geodesic transport, typed execution, and boundary measurement in one raw-token-to-answer pipeline.

3 Method: From Raw Tokens to Executable Geometry

3.1 Conceptual design

The geometry-first principle is meaningful only if it defines the complete computational route from raw input to an answer. That route has three responsibilities. First, it must infer relational structure directly from raw tokens without relying on an externally supplied graph, identity map, or externally specified symbolic program. Second, it must compose local relational evidence without losing entity identity, operator state, or payload during propagation. Third, it must distinguish a supported terminal state from a candidate that is merely reachable.

These requirements lead to three tightly coupled stages. *Geometry formation* constructs the persistent information field Ψ , computes pairwise observables, and induces multiple directed world geometries. *Geometric execution* assigns geodesic action to the induced geometry and performs GeoTransport, moving typed states while preserving identity, payload, and memory under admissibility constraints. Finally, *boundary adjudication* accepts a candidate only when the transported state satisfies the required propagation, memory, payload, and terminal constraints. Geometry is therefore part of the inference mechanism itself rather than an auxiliary representation: every stage produces the state or constraint required by the next.

Table 2 summarizes how the principal challenges of long-horizon relational reasoning motivate the architectural components of ERGT. The terminology is inspired by geometric and physical concepts but is used strictly in an operational sense. The persistent information field Ψ stores token-level information, world geometries represent directed relational structure, geodesic action defines traversal cost, the propagation cone enforces admissibility, GeoTransport propagates typed states, a geodesic corresponds to the least-cost admissible path, and boundary measurement performs the final acceptance test.

These components are constitutive design choices rather than claims of literal physical law. Their role is directly testable. The governing equations define their behavior, numerical invariants monitor their stability during optimization, and frozen interventions evaluate whether the trained model

Table 2: Conceptual mapping from design requirements to architectural components.

Problem requirement	Design principle	Architecture component
Preserve token identity and content	Persistent state from raw input	Persistent information field Ψ
Represent complementary relational views	Complementary directed geometries	Eight directed world geometries
Prevent invalid propagation	Explicit admissibility constraints	Finite-speed propagation cone
Preserve operators and payload	Type-preserving state propagation	GeoTransport and payload registers
Compose long relational paths	Dynamic programming over geodesic action	Min-plus geodesic closure
Reject unsupported solutions	Terminal consistency constraints	Boundary measurement and abstention

depends on them during inference. Continuous neural states construct the persistent information field and local relational evidence, whereas admissibility, typed-state preservation, memory, and terminal contracts determine whether those local computations constitute a valid reasoning path. Consequently, a high prediction score cannot override an invalid intermediate computation.

The resulting inference pipeline is

$$\begin{aligned} \text{raw tokens} &\longrightarrow \Psi \longrightarrow \text{pairwise observables} \longrightarrow \text{world geometries} \\ &\longrightarrow \text{GeoTransport} \longrightarrow \text{geodesic closure} \longrightarrow \text{boundary measurement}. \end{aligned}$$

No QKV self-attention, Transformer encoder, externally supplied graph, target compiler, or generic execution engine participates in this native inference path. Instead, relational structure, admissible propagation, typed transport, and boundary acceptance all emerge directly from the persistent information field constructed from the same raw input sequence.

3.2 Raw input and the persistent Ψ field

The construction begins with a representation that is sufficiently expressive to support relational reasoning without requiring a supplied graph or an external identity map. Semantic content alone is insufficient: repeated or similar tokens must remain distinguishable, yet identity cannot be introduced through an external pointer to the answer. The persistent information field therefore integrates semantic content, positional information, and an identity fibre, all computed from the same raw token. Every subsequent geometric quantity is derived from this shared initial representation.

For token i , the raw token identifier is mapped to a trainable semantic representation x_i and a collision-audited identity fibre u_i . Both are constructed internally by ERGT from the same public raw-token tensors. The initial persistent information field is

$$\Psi_i^{(0)} = f_\psi(x_i + p_i + \gamma f_{\text{id}}(u_i)), \quad (1)$$

where p_i denotes the positional representation and γ controls the contribution of the identity fibre. The identity component influences both the initial field and the pairwise observables derived from it while providing no external pointer, graph edge, condition label, or answer channel. At the public input boundary, the model receives only raw token identifiers together with a padding mask. Equation 1 therefore captures the central field-construction principle of ERGT: all subsequent relational geometry must emerge from pairwise observables of the same persistent information field available at the raw-input boundary.

3.3 Field observables and adhesion

The persistent information field must next determine which token pairs can support a relation. Rather than compressing that decision into a single unconstrained edge logit, ERGT exposes several bounded observables: coherence, reconstructibility, local compatibility, order, and agreement with the bulk-boundary view. Together these observables transform the persistent information field into candidate relational support while preserving the contribution of each factor to the final decision.

The persistent information field induces the following bounded observables:

$$\rho_i = d^{-1} \|\Psi_i\|_2^2, \quad (2)$$

$$C_{ij} = \frac{1}{2} [1 + \langle f_C(\Psi_i), g_C(\Psi_j) \rangle], \quad (3)$$

$$R_{ij} = \sigma(2 \langle f_R(\Psi_i), g_R(\Psi_j) \rangle), \quad (4)$$

$$G_{ij} = \exp(-|\rho_i - \rho_j|), \quad (5)$$

$$B_{ij} = \exp \left[-\frac{\delta_i^{\text{bulk}} + \delta_j^{\text{bulk}}}{2} \right]. \quad (6)$$

Here, C measures coherence, R reconstructibility, G density-gradient compatibility, and B bulk-boundary agreement. Local relational entropy S_i defines the order term $O_i = 1 - S_i$, while stability is measured across successive field updates.

The observables are combined into an adhesion field through a geometric mean in logarithmic space,

$$\Phi_{ij} = \exp \left(\frac{1}{6} \sum_{k=1}^6 \log(z_{ij}^{(k)} + \epsilon) \right), \quad z^{(k)} \in \{C, R, \text{stability}, G, O, B\}, \quad (7)$$

where each observable contributes multiplicatively to the final adhesion score. Consequently, a single weak factor cannot be compensated for by several strong ones, preserving the interpretability of the resulting edge support. The adhesion field Φ is a model-specific edge-formation mechanism rather than a universal principle of information processing.

3.4 Constitutive world geometries

The adhesion field establishes relational support, but it does not define a single universal notion of proximity. Locality, reconstructibility, long-range bridges, density flow, and boundary compatibility may each favor different directed relations. Collapsing these signals into a single geometry would discard their complementary structure. ERGT therefore induces eight constitutive world geometries from the shared persistent information field and preserves their geometric potentials until a typed computational role requires them. Table 3 summarizes the resulting world geometries.

For world geometry w , the directed edge state evolves according to

$$E_{wij}^{(t+1)} = \beta E_{wij}^{(t)} + (1 - \beta) \Phi_{ij}^{(t)} \Omega_{wij}^{(t)}, \quad \ell_{wij} = -\log(E_{wij} + \epsilon). \quad (8)$$

Typed computational roles define a routing mechanism over these distinct world geometries. Let $\bar{\Psi}$ denote the masked mean persistent information field, let r denote a typed role, let π_{rw} denote a learned role-world prior, and let $\mathcal{W}_r$ denote the world geometries authorized for role r . Then

$$\alpha_{rw} = \text{softmax}_{w \in \mathcal{W}_r} \left([W_{\text{route}} \bar{\Psi}]_{rw} + \pi_{rw} \right), \quad (9)$$

$$H_r = \sum_{w \in \mathcal{W}_r} \alpha_{rw} A_w(H_w), \quad (10)$$

Table 3: Operational world geometries induced from the persistent information field. These are constitutive geometric functions rather than externally supplied graphs.

World	Primary relational bias
W_0	locality, density-gradient compatibility, stability
W_1	coherence, identity overlap, local order
W_2	reconstructibility and boundary compatibility
W_3	stability, identity overlap, locality
W_4	long-range bridges, coherence, identity support, reconstructibility
W_5	density contrast and stability
W_6	novelty and long-range bridge support
W_7	source-sink density flow and coherence

where A_w adapts the representation produced by world geometry w . A world geometry is therefore not an independent predictor but one directed geometric view through which typed states may propagate. Different world geometries need not contribute equally, nor possess unique human-readable interpretations. The testable claim is instead functional dependence on their composition: under frozen interventions, no individual world geometry can reproduce the complete inference process.

3.5 Finite-speed admissibility and geodesic action

A learned edge potential indicates where a relation is plausible, but it does not by itself make that relation executable. Allowing every nonzero affinity to propagate would create dense shortcut paths capable of bypassing long relational chains. Each induced world geometry is therefore transformed into a sparse admissible domain, and every admissible transition is assigned a non-negative geodesic action. Admissibility determines *whether* information may propagate, whereas geodesic action determines the traversal cost of every admissible local transition.

Sparse support together with a finite propagation condition defines the admissible edges:

$$\mathcal{A}_{wij} = \mathbf{1}[j \in \text{topk}(E_{wi\cdot})] \mathbf{1}[\ell_{wij} \leq v_i \Delta \tau_{ij}]. \quad (11)$$

The geodesic action is

$$S_{wij} = \ell_{wij} + \lambda_b(1 - B_{ij}) + \lambda_s(1 - \text{stability}_{ij}) + \lambda_c \text{coneViolation}_{ij}. \quad (12)$$

The geodesic action defines a learned non-negative traversal cost. The finite-speed propagation cone is an operational computational constraint: edges whose effective traversal cost exceeds the available discrete propagation interval are rejected. It is introduced solely to enforce admissibility and carries no implication of a literal relativistic interpretation.

3.6 GeoTransport and geometric memory

Geodesic action becomes useful only if information can propagate along the selected path while preserving the computational state required for later reasoning. GeoTransport provides this state evolution. Its propagation mask is inherited directly from the admissibility constraints, and its transport weights are determined by geodesic action. Consequently, an edge rejected during geometric execution cannot reappear during transport. Persistent momentum, edge state, and transported payload preserve the information required by subsequent world geometries and boundary measurement.

Transport is induced by geodesic action over admissible neighbors:

$$T_{wij} = \text{softmax}_{j:A_{wij}=1} \left(-\frac{S_{wij}}{\tau} \right), \quad (13)$$

$$M_{wi} = \sum_j T_{wij} V_w(\Psi_j), \quad (14)$$

$$\Pi_i^{(t+1)} = \mu \Pi_i^{(t)} + (1 - \mu) F(\Psi_i^{(t)}, M_i^{(t)}), \quad (15)$$

$$\Psi_i^{(t+1)} = \text{Norm} \left(\Psi_i^{(t)} + \Pi_i^{(t+1)} \right). \quad (16)$$

Geometric memory is the path-dependent computational state retained in the evolving edge state E , momentum state Π , and transported payload. This memory influences subsequent world geometries and boundary measurement, but it provides no direct shortcut to the final decision. GeoTransport is therefore a masked, geodesic-action-weighted transport operator that keeps typed states attached to the executable path along which they propagate, rather than performing attention-weighted value mixing.

3.7 Typed event support

Geometry determines where state may propagate and the cost of each transition, but it does not by itself determine the relational operator carried by an event. That relational semantics must remain sufficiently discrete to prevent one relation type from being replaced by another simply because it lies on a lower-cost path. Typed event support therefore binds the predicted source, relation, and target roles to the propagated state, while the induced geometry retains control over admissibility and route selection. This forms the interface between learned token interpretation and executable geometric reasoning without relying on an externally supplied relational graph.

Predicted source, relation, and target distributions define the typed event support

$$Q_{ijr} = \sum_e p(i \mid e) p(r \mid \Psi_e^{(0)}) p(j \mid e). \quad (17)$$

The typed event support provides local relational proposals to the geometric program. The induced geometry determines whether a proposal is admissible and, if so, the cost of incorporating it into an executable reasoning path, while the discrete relation type carried by Q remains unchanged throughout transport. This separation prevents route optimization from silently altering the semantic operator associated with the propagated state.

3.8 Geodesic closure on the token–world product space

The preceding stages define admissible local transitions. To support long-horizon reasoning, these transitions must be composed without delegating the final decision to a fixed-depth classifier. ERGT achieves this through dynamic programming over the discrete product space of token locations and world geometries. One coordinate identifies the current token location, while the other specifies the directed world geometry under which the transition is evaluated. Min-plus closure then recursively composes the same local admissibility and geodesic-action rules into executable paths spanning multiple propagation steps.

The solver state is the discrete product (i, w) of token location and world geometry. For candidate c , the min-plus field is initialized at its registered source:

$$D_c^{(0)}(i, w) = \begin{cases} 0, & (i, w) \text{ is an admissible source,} \\ +\infty, & \text{otherwise,} \end{cases} \quad (18)$$

$$D_c^{(t+1)}(j, w') = \min \left\{ D_c^{(t)}(j, w'), D_c^{(t)}(i, w) + A[(i, w) \rightarrow (j, w')] \right\}, \quad (19)$$

where $A[(i, w) \rightarrow (j, w')]$ denotes the learned geodesic action associated with an admissible transition.

Support is propagated through a max-min widest-path formulation, thereby preventing repeated probability attenuation from suppressing otherwise valid long reasoning paths.

Closure depth is selected to exceed the target reasoning horizon and is capped at 36 propagation steps in the experimental configuration. A label-free admissibility domain enables geodesic action to propagate over long raw-token sequences without introducing an externally supplied answer graph.

Throughout this work, a *geodesic* denotes the admissible path with minimum accumulated action under the learned discrete geometry.

3.9 Typed payload and boundary measurement

A least-action path establishes reachability, but reachability alone is insufficient for a valid answer. The arriving state must also encode the operator sequence required by the candidate, preserve sufficient transported payload, and satisfy the registered terminal boundary conditions. The final stage couples geodesic optimization with typed register execution and determines, through boundary measurement, whether the assembled evidence satisfies the complete execution contract. Fail-closed behavior therefore becomes an intrinsic property of the inference mechanism rather than a post-processing convention.

Along an admissible event, the register state and transported payload evolve according to

$$r' = \text{applyOperator}(r, r_e), \quad (20)$$

$$q^{(t+1)}(j, r') = \max_e q^{(t)}(i_e, r) \tau_e m_e, \quad (21)$$

$$\Delta^{(t+1)}(j, r') = \min_e \left[\Delta^{(t)}(i_e, r) + \delta_e \right]. \quad (22)$$

Before checkpoint eligibility, two normalization contracts are verified:

$$\varepsilon_{\text{typed}} = \max_e \left| \sum_r p_e(r) - p_e(\text{present}) \right|, \quad (23)$$

$$\varepsilon_{\text{transport}} = \max_{w, i \text{ valid}} \left| \sum_j T_{wij} - 1 \right|. \quad (24)$$

The first enforces consistency between relation-type mass and event-presence mass, whereas the second guarantees that every valid transport row remains normalized.

For candidate c , let A_c denote its accumulated geodesic action, m_c its terminal payload mass, and δ_c its boundary deficit. Its validity indicator is

$$V_c = \mathbf{1}[A_c < \infty] \mathbf{1}[m_c \geq m_{\min}] \mathbf{1}[\delta_c \leq \delta_{\max}], \quad (25)$$

where A_c is evaluated from the typed register associated with that candidate, so an incorrect register cannot satisfy the validity criterion. Propagation admissibility has already been enforced during construction of the finite-action geodesic paths.

The final decision is

$$\hat{c} = \arg \min_{c: V_c=1} A_c \quad \text{only if } A_{(2)} - A_{(1)} \geq \eta; \quad \text{otherwise abstain.} \quad (26)$$

Boundary measurement therefore verifies the final typed register, transported payload, conservation constraints, and boundary deficit before accepting a candidate. Multiple unresolved candidates, the absence of any valid candidate, or an insufficient geodesic margin all lead to abstention. The decision is performed natively within the induced geometry rather than being delegated to an external graph execution engine.

3.10 Training objective and teacher-free inference

Training must reconcile differentiable field construction with hard execution contracts. The persistent information field requires sufficient supervision to establish the native geometric reasoning pipeline, yet the reported inference must not depend on teacher guidance or auxiliary execution shortcuts. Accordingly, intermediate losses are grouped by their architectural roles, whereas checkpoint selection is separated from the subsequent frozen teacher-free evaluation.

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{answer}} + 2a(t)\mathcal{L}_{\text{structured}} + 2a(t)\mathcal{L}_{\text{execution}} + 2a(t)\mathcal{L}_{\text{causal}} + 2a(t)\mathcal{L}_{\text{margin}} + 0.5\mathcal{L}_{\text{geometry}}. \quad (27)$$

Here $a(t)$ denotes the teacher schedule. Structured losses supervise entity identity, query role, event type, endpoints, operator, boundary, and edge construction. Execution losses supervise geodesic action, propagation admissibility, transport, boundary deficit, and terminal payload. Geometry losses include bulk–boundary reconstruction, conservation, world utilization, and geometric diversity.

A checkpoint becomes eligible only after satisfying independent field-readiness criteria. The teacher weight is then reduced to zero, all model parameters are frozen, and two additional validation windows are completed without any parameter updates. Consequently, every result reported in this paper is obtained under teacher-free inference, although supervised learning is employed during training.

4 Experimental Design

The experimental protocol is designed to mirror the computational pipeline introduced in Section ?? . It evaluates three successive questions. First, can raw-token input induce a stable executable geometry? Second, after training has converged, does the frozen computational mechanism continue to solve increasingly long relational chains? Third, do the reported predictions genuinely depend on the geometric operations claimed by the model? A separate unsupported-input protocol evaluates whether the boundary solver correctly abstains when no admissible solution exists. Because these questions address different aspects of the proposed mechanism, each is evaluated through a separate evidence panel rather than a single aggregate accuracy score.

The evidence panels are also isolated during model selection. Development data are used exclusively to determine the strengthened Transformer baseline. Native-readiness data identify the earliest stable ERGTcheckpoint that satisfies the registered execution contracts. The long-horizon evaluation cohorts remain unseen until both selection procedures are complete. All ablation studies and geometric interventions are then performed only on this single frozen checkpoint. Table 4 summarizes the resulting inferential structure.

Table 4: Inferential structure of the registered experimental protocol.

Question	Evidence panel	Supported conclusion
Has the executable inference path been learned?	Training set, raw-disjoint validation, and native execution contracts	Model convergence and, for ERGT, readiness of the executable geometric pipeline.
Does computation generalize to longer reasoning horizons?	Frozen matched evaluation cohorts beyond the training horizon	Long-horizon reasoning performance and paired model comparison.
Does the model fail closed?	Controlled unsupported-input protocol	Reliable abstention when no admissible solution exists.
Are geometric components functionally necessary?	Frozen-checkpoint interventions, removals, corruption, and single-world controls	Causal attribution to the intervened geometric mechanism on eligible solved instances.
How does the induced geometry evolve with reasoning horizon?	Detached spectral, curvature, and world-diversity observers	Descriptive geometric characterization only, without causal attribution of performance.

4.1 Benchmark and counterfactual evaluation cohorts

The benchmark is designed to isolate relational reasoning from open-domain language knowledge. Each example serializes entities, relation events, a query, two candidate answers, and distractor tokens into a single raw-token sequence. This serialized sequence constitutes the complete input presented to both models. A latent graph exists only inside the data generator to produce matched relational instances; graph adjacency, edge indices, entity identifiers, and candidate paths are never exposed to either model.

The fundamental evaluation unit is an answer-reversing counterfactual pair. Both members of a pair share graph topology, edge count, reasoning depth, operator sequence, and lexical statistics, while one registered intervention changes the unique valid answer. Individual scenarios independently manipulate geodesic action, propagation admissibility, boundary deficit, transported payload, terminal mass, source memory, and combinations of these mechanisms over increasingly long reasoning chains. Consequently, successful performance requires identifying the governing execution condition rather than memorizing surface patterns or candidate positions.

Training comprises 28 counterfactual pair groups at each reasoning depth from 1 to 8 hops, yielding 224 supported pair groups together with 28 controlled unsupported groups. Independent native-readiness data cover reasoning depths from 2 to 8 hops. Frozen evaluation is performed at reasoning horizons of 4, 8, 12, 16, 24, and 32 hops. Separate context-length panels evaluate serialized sequences containing 128, 256, 512, and 768 tokens while keeping the underlying reasoning task fixed. Unsupported examples are evaluated at 4, 16, and 32 hops. The horizon panels therefore isolate reasoning depth, whereas the context panels isolate sequence length under an otherwise unchanged reasoning problem. This controlled synthetic benchmark is intended to evaluate the proposed computational mechanism and should not be interpreted as a measure of open-domain language understanding.

4.2 Compared inference pipelines

The comparison is defined at the external task boundary rather than by forcing the two architectures to share an internal reasoning mechanism. Both models receive exactly the same serialized raw-token sequence and padding mask, are trained on the registered 1–8-hop benchmark under fixed optimization protocols, and predict one of three outcomes: candidate A, candidate B, or abstention. The Transformer baseline is strengthened through the same training protocol, but no ERGT-derived geometric structure

participates in its inference pipeline. The comparison therefore evaluates two complete end-to-end inference pipelines operating on identical raw input rather than comparing ERGT with a Transformer augmented by the ERGT execution mechanism.

The direct baseline is implemented as a standard PyTorch Transformer encoder consisting of token embeddings, sinusoidal positional encoding, multi-head QKV self-attention, GELU feed-forward blocks, layer normalization, and a three-class prediction head. Its locked configuration uses hidden width 64, four attention heads, four encoder layers, feed-forward width 192, maximum context length 768, and zero dropout. The resulting model contains 205,634 inference-active parameters, compared with 196,816 for ERGT, a difference of 4.48%. Auxiliary supervision is used only during training and never as an inference input. The baseline receives no graph structure, geometric state, symbolic program, execution engine, identity side channel, or other task-specific information beyond the serialized raw input.

Native ERGT uses hidden width 56, a Ψ rank of 20, eight induced world geometries, three field updates, sparse top-20 relational support, and at most 36 geodesic-closure iterations. Starting from the same serialized input and padding mask, it must construct identity, relational support, executable world geometries, typed transport, and the final boundary decision entirely within its native inference pipeline. No learned answer MLP bypasses this execution path. The close parameter counts reduce a simple capacity-based explanation of performance differences, although they do not imply identical computational cost. Optimizer steps, execution time, and peak CUDA memory are therefore reported rather than artificially matched.

4.3 Locked training protocol and checkpoint selection

The Transformer training protocol was finalized on development data before any 12–32-hop evaluation cohort was accessed. Training begins with a one-hop warm-up using batch size 8 and learning rate 7×10^{-4} . After two successful raw-disjoint validation windows, the curriculum may progress through fixed 1–2, 1–4, and finally 1–8-hop training stages. Once established, the architecture, optimizer, curriculum-transition rules, and optimization budget remain fixed throughout all confirmatory experiments. Any missed curriculum transition is recorded rather than adjusted using information from the final evaluation cohorts.

The confirmatory `four_seed_confirmation` protocol trains both models independently from scratch. Optimization seeds 17117, 18313, 19501, and 20707 are paired with independent data-generation seeds 21101, 22307, 23509, and 24733. All serialized datasets are collision-audited and remain strictly disjoint across training, development, validation, and final evaluation. Long-horizon, context-length, unsupported, intervention, and observer panels are excluded entirely from checkpoint selection.

During confirmation, the Transformer follows the locked warm-up and curriculum schedule for a maximum of 1400 optimization steps. ERGT is trained on the registered 1–8-hop benchmark using batch size 8, AdamW with learning rate 2.5×10^{-3} , and at most 1800 optimization steps. Teacher guidance is used only during training. The first checkpoint that successfully satisfies two consecutive 2–8-hop field-readiness windows is immediately frozen and subsequently evaluated over two additional teacher-free validation windows without further parameter updates. Checkpoint selection therefore depends exclusively on stable short- and intermediate-horizon execution, never on the long-horizon evaluation results.

4.4 Registered outcomes and interventions

Per-example accuracy measures individual predictions, whereas counterfactual-pair exactness requires both members of an answer-reversing pair to be predicted correctly. Model-specific convergence

requires at least 0.90 supported accuracy on both the training ID split and the untouched raw-disjoint validation split. A secondary matched-convergence analysis is restricted to optimization/data seed pairs for which both models satisfy this criterion. Failure of one model limits only joint-convergence claims and does not affect the independent evaluation of the other.

The preregistered primary comparison is the 32-hop evaluation panel. The confirmatory claim requires native accuracy of at least 0.90, an absolute advantage of at least 0.35 over the direct Transformer, a hierarchical bootstrap confidence interval excluding zero, and a paired exact McNemar test with $p < 0.05$. Counterfactual pairs define the paired bootstrap clusters and are nested within data seeds. Bootstrap confidence intervals follow Efron and Tibshirani [6], whereas discordant outcomes are evaluated using McNemar’s exact test [14]. Intermediate reasoning horizons, context-length panels, counterfactual-pair exactness, and unsupported rejection serve as supporting analyses and are never used for checkpoint selection.

Checkpoint eligibility for ERGTis governed by stricter criteria than answer accuracy alone. Measurement accuracy must reach 0.99; geodesic-action MAE must not exceed 0.12; event recall, precision, path coverage, and chain exactness must reach 0.99, 0.98, 0.99, and 0.99, respectively; geodesic closure must not degrade; hard overflow must remain zero; the conservation residual must not exceed 10^{-4} ; every registered group must reach exposure 8; and two consecutive readiness windows must be satisfied. Batch, padding, identity, and raw-input invariance are recorded together with these execution contracts. All registered outcomes are reported regardless of whether a claim-specific criterion is satisfied.

Mechanism attribution is performed using the selected checkpoint without retraining. Each intervention modifies a single inference operation and is evaluated only on counterfactual pairs solved by the intact model. Targeted interventions independently remove geodesic action, finite-speed admissibility, typed transport, boundary deficit, terminal payload, or source-conditioned memory. Global controls randomize or shuffle the induced geometry, bypass the conditioned world-geometry router, or retain only a single world geometry. The preregistered minimum targeted effect is an absolute accuracy reduction of 0.20. Formal mechanism evaluation additionally requires intact accuracy of at least 0.80 together with at least two eligible counterfactual pairs. Conditional intervention effects are still reported below this panel-wide threshold, although the corresponding mechanistic claims are restricted accordingly.

Curvature, graph gradient, Laplacian spectrum, spectral entropy, effective rank, spectral gap, and world diversity are treated differently. These quantities characterize the frozen induced geometry without participating in the executable inference path and therefore cannot influence predictions. Their role is descriptive rather than causal, providing structural characterization of the learned geometry and guidance for future analyses rather than explaining current predictive performance.

5 Results

The results are organized to follow the execution of a single learned inference program. We first verify that the native computation converges and can be frozen without violating its registered execution contracts. The resulting frozen checkpoints are then evaluated under progressively greater relational horizons and longer input contexts before being subjected to unsupported inputs, mechanism interventions, and detached geometric observation. Consequently, convergence, extrapolation, and mechanism attribution remain distinct empirical questions while referring to the same executable inference path.

Table 5: Training and raw-disjoint validation accuracy across independent optimization/data-seed pairs. Reported values are obtained from fresh confirmatory runs.

Seed pair	ERGTtrain	ERGTval.	TRANSFORMERtrain	TRANSFORMERval.	Freeze step
17117/21101	1.000	1.000	1.000	0.510	700
18313/22307	1.000	1.000	1.000	0.459	700
19501/23509	1.000	1.000	1.000	0.837	1000
20707/24733	1.000	1.000	1.000	0.449	1000

5.1 Convergence and checkpoint integrity

The long-horizon evaluation begins with the native computation that is subsequently extrapolated. As shown in Table 5, native ERGT achieved perfect accuracy on both the training split and the raw-disjoint validation split for all four independent optimization/data-seed pairs. The complete inference pipeline—from persistent-field construction through world geometries, geodesic execution, GeoTransport, and boundary measurement—therefore converged consistently across independent initializations and previously unseen data. The strengthened direct Transformer likewise reached perfect training accuracy in every run, whereas its validation accuracy ranged from 0.449 to 0.837. These results characterize the registered baseline and do not affect the independent convergence result of the native ERGT computation.

Answer accuracy alone was insufficient for checkpoint selection. The first checkpoint satisfying two consecutive native-readiness windows was frozen only after every registered 2–8-hop group had reached the required exposure. The resulting freeze steps were 700, 700, 1000, and 1000, while the corresponding minimum group exposures were 11, 11, 15, and 15, exceeding the preregistered minimum of 8 in every case. No registered group remained unseen, and no optimizer updates were applied after checkpoint freezing.

The registered execution contracts yielded the same conclusion at finer granularity. All 56 evaluations of the 14 native stability invariants were satisfied. Event-chain exactness, geodesic path coverage, geodesic-closure integrity, event precision, and event recall each reached 1.000; hard overflow remained zero; and the maximum typed-payload conservation residual was 2.38×10^{-7} . Batch-size invariance, padding invariance, raw-input identity construction, and raw-input parity were likewise satisfied. Collectively, these results indicate that the selected checkpoints represent stable executions of the intended native inference program rather than artifacts of checkpoint drift, external identity signals, or serialization effects.

5.2 Frozen generalization across horizon and context

The selected checkpoints were then frozen for all subsequent generalization experiments. Because training was restricted to the registered 1–8-hop family and no final evaluation panel participated in checkpoint selection, the following results measure extrapolation rather than continued adaptation. As summarized in Table 6, native ERGT maintained accuracy above 0.90 across every registered reasoning horizon and achieved a mean accuracy of 0.9554 at 32 hops, whereas the strengthened direct Transformer remained close to chance throughout the evaluation.

The aggregate results also reveal the variability across independent seed pairs. At the registered 32-hop evaluation, native ERGT achieved accuracies of 0.982, 1.000, 1.000, and 0.839 across the four optimization/data-seed pairs. The corresponding exactness on the answer-reversing counterfactual pairs was 0.964, 1.000, 1.000, and 0.750, indicating that the overall mean is not dominated by a single training run.

The pooled accuracies at the primary evaluation horizon were

Table 6: Mean accuracy across four independent optimization/data-seed pairs. Training covers reasoning horizons from 1 to 8 hops.

Hops	Native ERGT	Direct Transformer	Difference
4	1.0000	0.4821	0.5179
8	1.0000	0.5045	0.4955
12	0.9821	0.5000	0.4821
16	0.9866	0.5000	0.4866
24	0.9196	0.5000	0.4196
32	0.9554	0.5000	0.4554

Table 7: Mean correct rejection rate on controlled unsupported examples.

Hops	Native ERGT	Direct Transformer
4	1.0000	0.0000
16	1.0000	0.0000
32	1.0000	0.0000

$$\text{Acc}_{\text{ERGT}} = 0.955357, \quad \text{Acc}_{\text{TRANSFORMER}} = 0.500000.$$

The resulting paired difference of 0.455357 exceeds the preregistered margin of 0.35, while the native mean remains above its preregistered accuracy threshold of 0.90. A hierarchical bootstrap over data seeds and counterfactual pairs produced a 95% confidence interval of $[0.375, 0.500]$, excluding zero throughout the interval. The paired exact McNemar test yielded

$$p_{\text{McNemar}} = 9.86 \times 10^{-32}.$$

Among the 224 evaluated examples (112 counterfactual pairs), 104 were answered correctly only by native ERGT, whereas none were answered correctly only by the direct Transformer. This asymmetry is consistent with both the bootstrap confidence interval and the paired exact test. Within the registered benchmark, the frozen native geometric inference path therefore preserves substantially higher accuracy than the direct Transformer as relational depth increases well beyond the training horizon.

The context-length panels separate robustness to longer serialized inputs from extrapolation across relational depth. At the fixed 4-hop horizon, mean native ERGT accuracy remained 1.000, 1.000, 1.000, and 0.963 for contexts of 128, 256, 512, and 768 tokens, respectively, whereas the corresponding Transformer accuracies were 0.500, 0.506, 0.519, and 0.500. At the more demanding 24-hop horizon with contexts of 512 and 768 tokens, native ERGT achieved mean accuracies of 0.931 and 0.988, compared with 0.519 and 0.531 for the direct Transformer. These context panels therefore indicate that the observed long-horizon advantage is maintained under the registered longer serializations, while making no claim regarding general long-context language modeling beyond the present benchmark.

5.3 Fail-closed behavior and geometric attribution

The next question is not how often the model produces an answer, but whether it answers only when its own execution contracts are satisfied. On controlled examples for which neither candidate possesses admissible transported support, ERGT rejected every case at 4, 16, and 32 hops across all optimization/data seed pairs (Table 7), whereas the direct Transformer never abstained.

Table 8: Frozen mechanism interventions. Full and intervention accuracies are computed only on eligible counterfactual pairs correctly solved by the intact model. “Formal seeds” additionally satisfy the preregistered panel-wide 0.80 intact-model evaluability criterion.

Removed mechanism	Full	Intervention	Formal seeds
Geodesic action	1.000	0.000	4/4
Finite-speed cone	1.000	0.000	4/4
GeoTransport	1.000	0.000	4/4
Terminal mass	1.000	0.000	4/4
Source-conditioned memory	1.000	0.031	4/4
Boundary deficit	1.000	0.000	2/4

Within the registered benchmark, this constitutes fail-closed behavior: absence of a valid executable geometric solution results in abstention rather than a forced prediction. This result is specific to the controlled benchmark and should not be interpreted as an evaluation of open-domain hallucination.

The same frozen checkpoints were then used to determine whether successful predictions actually depend on the proposed geometric computation. Table 8 summarizes scenario-matched interventions evaluated only on counterfactual pairs correctly solved by the intact model, so each reported decrease measures the loss of an established capability rather than overall task performance. Geodesic action ranks admissible paths, the finite-speed cone prevents invalid propagation, GeoTransport preserves typed state, terminal mass verifies successful arrival, source-conditioned memory retains path history, and boundary deficit enforces terminal consistency.

Removing geodesic action, finite-speed admissibility, GeoTransport, or terminal mass reduced eligible accuracy from 1.000 to 0.000 across all four optimization/data-seed pairs. Removing source-conditioned memory reduced the corresponding accuracy to 0.031.

Removing the boundary-deficit mechanism likewise reduced eligible solved-pair accuracy from 1.000 to 0.000 in every optimization/data-seed pair. However, only two of the four runs satisfied the preregistered evaluability criterion because the intact boundary panel reached 0.750 in the remaining two runs, below the conservative 0.80 threshold. Accordingly, the conditional effect is observed consistently across all runs, whereas the formal preregistered claim is restricted to the two eligible seed pairs.

Global control interventions lead to the same conclusion. Randomizing the induced geometry reduced eligible accuracy by 0.646, cross-example geometry shuffling reduced it by 0.597, and bypassing the conditioned world router reduced it by 0.382. Restricting inference to any individual world geometry (W_0 – W_7) yielded zero eligible accuracy.

Taken together, these interventions indicate that successful inference depends on the learned organization of admissible costs, transport constraints, and their composition across multiple world geometries, rather than on any single world or isolated computational component.

5.4 Detached geometric observers

The detached geometric observers provide descriptive information about the learned geometry without participating in inference. Curvature and spectral statistics are computed entirely outside the executable reasoning path and are never available to the prediction mechanism. The detached CPU/float64 analysis processed all 26,176 induced graphs without missing values, numerical instability, sanitization, or CUDA eigensolver calls. Table 9 summarizes the resulting geometric statistics.

As relational depth increases from 4 to 32 hops, the effective spectral rank increases from 4.859 to 43.867, spectral entropy from 0.711 to 0.868, and world diversity from 0.1447 to 0.8052. At the same time, the spectral gap decreases from 0.1512 to 0.0115, and the mean absolute curvature gradient

Table 9: Detached geometric observers averaged across four independent optimization/data-seed pairs.

Observer	4 hops	32 hops
Mean curvature proxy	−0.1011	−0.0154
Negative-curvature fraction	0.8959	0.9479
Mean absolute curvature gradient	0.002424	0.000354
Spectral effective rank	4.859	43.867
Spectral entropy	0.711	0.868
Spectral gap	0.1512	0.0115
World diversity	0.1447	0.8052

decreases from 0.002424 to 0.000354, while the Forman-style curvature proxy remains predominantly negative throughout.

These observations suggest that the induced geometry becomes progressively more distributed across spectral modes and participating world geometries as reasoning depth increases. Because all observer statistics are computed outside the executable inference path, they provide only a structural description of the learned geometry and should not be interpreted as evidence that curvature or spectral properties themselves cause the observed generalization performance.

5.5 Computational cost

The proposed executable geometry incurs a higher computational cost than the direct Transformer baseline. Across the four confirmatory runs on the Colab runtime, mean training time was 1437.5 seconds for ERGT compared with 56.25 seconds for the direct Transformer, corresponding to a per-seed ratio between 21.3 and 29.8. Mean peak CUDA memory allocation was approximately 3.432 GiB for ERGT and 1.455 GiB for the direct Transformer, a ratio of 2.36.

These measurements are reported for transparency rather than as evidence of computational efficiency. The experiments evaluate whether the proposed geometry enables a different inference mechanism, not whether it is the most resource-efficient implementation.

Taken together, the four-seed evaluation forms a consistent empirical chain. The native raw-token-to- Ψ computation converges under the registered training protocol, remains stable after freezing, generalizes to substantially longer reasoning horizons, exhibits fail-closed behavior on unsupported examples, and loses its predictive capability when its principal geometric mechanisms are selectively removed. The reported performance gains are therefore accompanied by evidence that geodesic action, GeoTransport, source-conditioned memory, and terminal boundary constraints actively participate in inference, albeit at the additional computational cost reported above.

6 Discussion

The results support a consistent conclusion: within the registered benchmark, ERGT learns an executable inference path that extends from raw tokens to a constrained terminal decision. Rather than serving as an intermediate latent representation, the persistent information field Ψ provides the identity and relational evidence from which the model induces world geometries, performs GeoTransport, and evaluates terminal boundary conditions. This native computation converged across all four independent optimization/data seeds and remained stable after checkpoint freezing, demonstrating that the proposed geometric pipeline can be optimized end-to-end and executed as the model’s inference mechanism.

The frozen horizon experiments further show that this computation extends beyond the relational

depths observed during training. Because the 12–32-hop evaluation panels were excluded from checkpoint selection, the reported performance reflects extrapolation of a fixed computation rather than adaptation to the evaluation horizon. The accompanying context controls indicate that the observed gains are associated with deeper relational composition rather than simply longer input sequences, while the unsupported panels show that the same computation can abstain when no executable solution satisfies the registered terminal conditions.

The comparison with the strengthened direct Transformer defines the scope of this result. Both models receive identical serialized inputs, matched supervision, and similar inference-active parameter counts, but only ERGT constructs and executes an induced relational geometry. Within this benchmark, that explicit geometric execution generalizes more effectively across raw-disjoint validation and substantially longer reasoning horizons. This should be interpreted as an architectural result for the registered benchmark, not as a general limitation of Transformer models.

The intervention experiments provide complementary evidence that this performance depends on the proposed computation itself. Removing geodesic action, admissibility, GeoTransport, terminal verification, or source-conditioned memory consistently degraded the frozen model, while randomized geometry and single-world controls produced similar failures. Together, these results indicate that successful inference depends on the coordinated interaction of geometry, transport, memory, and boundary constraints rather than on an unrelated predictor.

The broader contribution of ERGT is therefore not simply the removal of self-attention. It is the construction of executable relational structure directly from raw tokens and the use of that structure as the inference mechanism itself. Instead of operating on an externally supplied graph or treating geometry as a post hoc explanation, ERGT derives relational structure internally and executes reasoning through admissibility, geodesic action, typed transport, and boundary measurement. The detached curvature and spectral observers complement this picture by describing how the induced geometry evolves with reasoning depth, although they remain outside the inference path and provide descriptive rather than causal evidence.

Overall, the evidence supports a bounded conclusion: within the registered relational benchmark, a persistent information field constructed from raw tokens can induce an executable geometry that converges reliably, supports auditable long-horizon reasoning, and operates through an explicitly defined fail-closed inference process.

7 Scope and Limitations

This study evaluates a bounded architectural hypothesis: that geometry induced directly from raw tokens can support long-horizon relational reasoning without externally supplied graphs or attention-based sequence mixing. Internal validity within the registered synthetic benchmark is supported by four independent optimization/data-seed pairs, raw-disjoint evaluation, frozen checkpoints, paired counterfactual design, invariant audits, and same-checkpoint interventions.

Whether these findings generalize to natural-language benchmarks, open-domain reasoning, substantially larger models, or alternative training objectives remains an open question. The strengthened direct Transformer baseline was locked before the confirmatory experiments but did not satisfy its development held-out criterion on the fresh evaluation data. Accordingly, the comparison supports an architectural advantage within the registered benchmark rather than a general superiority claim.

Training uses structured supervision, whereas inference is entirely teacher-free and requires neither an external compiler nor an execution engine. Boundary-deficit interventions satisfy the preregistered formal evaluability criterion in two of the four seed pairs, although the conditional intervention effect appears in all four runs. Curvature and spectral statistics remain detached descriptive observers, computational efficiency is not claimed, and fail-closed behavior is demonstrated only within the

registered unsupported protocol.

This work introduces ERGT, an attention-free architecture in which relational reasoning is performed through an executable geometry induced directly from raw token sequences. Instead of relying on self-attention or an externally supplied graph, the model constructs a persistent information field Ψ , induces multiple directed world geometries, propagates typed states through GeoTransport along admissible geodesics, and accepts an answer only after terminal boundary verification. The induced geometry is therefore the model’s native inference mechanism rather than a post hoc representation.

Across four independent optimization/data-seed pairs, the complete raw-token-to-boundary computation converged under the registered training protocol, remained stable after checkpoint freezing, and satisfied the registered execution contracts. Within the controlled relational benchmark, the frozen computation generalized well beyond the training horizon, demonstrated fail-closed behavior on unsupported examples, and consistently degraded under mechanism-specific interventions. These complementary results indicate that the reported predictions depend on the proposed executable geometry rather than on an independent prediction mechanism.

The contribution of this work is therefore not simply the removal of self-attention. It demonstrates that a persistent information field constructed from raw tokens can support an executable geometric reasoning process whose admissibility constraints, transport dynamics, and terminal verification remain active throughout inference. Although the present evidence is limited to a controlled relational benchmark, it establishes a concrete architectural basis for investigating executable information geometry as an alternative direction for long-horizon neural reasoning.

Statements and Declarations

Competing Interests

The authors declare that they have no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Author Contributions

Jalal Jafari: Conceptualization, Methodology, Software, Formal analysis, Writing – Original draft, Visualization. Nasibeh Aramideh: Methodology, Validation, Formal analysis, Writing – Review & Editing.

Data Availability

The complete implementation, synthetic benchmark generator, and experimental pipeline are publicly available at: <https://github.com/jalaljafari2009/ERGT-paper>

References

- [1] Shun-ichi Amari. *Information Geometry and Its Applications*. Springer, 2016.
- [2] Peter W. Battaglia, Jessica B. Hamrick, Victor Bapst, Alvaro Sanchez-Gonzalez, Vinicius Zambaldi, Mateusz Malinowski, Andrea Tacchetti, David Raposo, Adam Santoro, Ryan Faulkner, et al.

- Relational inductive biases, deep learning, and graph networks. *arXiv preprint arXiv:1806.01261*, 2018.
- [3] Beatrice Bevilacqua, Kyriacos Nikiforou, Borja Ibarz, Ioana Bica, Michela Paganini, Charles Blundell, Jovana Mitrovic, and Petar Veličković. Neural algorithmic reasoning with causal regularisation. In *Proceedings of the 40th International Conference on Machine Learning*, pages 2272–2288, 2023.
 - [4] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.
 - [5] Fan R. K. Chung. *Spectral Graph Theory*, volume 92. American Mathematical Society, 1997.
 - [6] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.
 - [7] Marc Finzi, Ke Alexander Wang, and Andrew G. Wilson. Simplifying hamiltonian and lagrangian neural networks via explicit constraints. In *Advances in Neural Information Processing Systems*, volume 33, pages 13880–13889, 2020.
 - [8] Robin Forman. Bochner’s method for cell complexes and combinatorial ricci curvature. *Discrete & Computational Geometry*, 29(3):323–374, 2003.
 - [9] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1263–1272, 2017.
 - [10] Samuel Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
 - [11] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.
 - [12] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. *arXiv preprint arXiv:2111.00396*, 2022.
 - [13] Thomas N. Kipf, Ethan Fetaya, Kuan-Chieh Wang, Max Welling, and Richard S. Zemel. Neural relational inference for interacting systems. In *Proceedings of the 35th International Conference on Machine Learning*, pages 2688–2697, 2018.
 - [14] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
 - [15] Michael Poli, Stefano Massaroli, Eric Nguyen, Daniel Y. Fu, Tri Dao, Stephen Baccus, Yoshua Bengio, Stefano Ermon, and Christopher Ré. Hyena hierarchy: Towards larger convolutional language models. In *Proceedings of the 40th International Conference on Machine Learning*, pages 28043–28078, 2023.
 - [16] Jake Topping, Francesco Di Giovanni, Benjamin P. Chamberlain, Xiaowen Dong, and Michael M. Bronstein. Understanding over-squashing and bottlenecks on graphs via curvature. *arXiv preprint arXiv:2111.14522*, 2022.
 - [17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

- [18] Petar Veličković, Adrià Puigdomènech Badia, David Budden, Razvan Pascanu, Andrea Banino, Misha Dashevskiy, Raia Hadsell, and Charles Blundell. The clrs algorithmic reasoning benchmark. In *Proceedings of the 39th International Conference on Machine Learning*, pages 22084–22102, 2022.
- [19] Zhaocheng Zhu, Zuobai Zhang, Louis-Pascal Xhonneux, and Jian Tang. Neural bellman–ford networks: A general graph neural network framework for link prediction. In *Advances in Neural Information Processing Systems*, volume 34, pages 29476–29490, 2021.

A Operational Physics Glossary

Table 10 summarizes the operational meaning of the physics-inspired terminology used throughout this paper. These terms denote implemented computational mechanisms within ERGTand should not be interpreted as literal physical objects or laws.

Table 10: Operational interpretation of physics-inspired terminology.

Term	Implemented meaning	Scope
persistent information field Ψ	Persistent token state built from semantics, position, and the internal identity fibre.	Computational latent state; not a physical field.
Density ρ	Mean squared field amplitude used during geometry construction.	Internal observable; not physical density.
Adhesion field Φ	Multiplicative edge-support score combining the six bounded observables.	Model-specific edge-formation rule.
World geometry	One of eight induced directed relational geometries derived from Ψ .	Computational geometry; not a physical world.
Geodesic action	Learned non-negative traversal cost on admissible edges.	Discrete path cost; not a physical action.
Finite-speed propagation cone	Admissibility rule limiting executable propagation.	Computational constraint; not relativity.
GeoTransport	Geometry-guided transport of typed states, momentum, and payload.	Transport operator; not QKV attention.
Geometric memory	Path-dependent state retained during transport.	Execution memory; not biological memory.
Typed event support	Discrete source–relation–target support predicted from Ψ .	Internal relation representation.
Product space	Discrete token–world state space for geodesic closure.	Finite computational state space.
Geodesic	Minimum-action admissible path computed by min-plus closure.	Shortest computational path.
Boundary measurement	Final verification of typed state, payload, and terminal constraints.	Decision rule for acceptance or abstention.
Boundary deficit	Mismatch between transported state and terminal constraints.	Decision observable.

Term	Implemented meaning	Scope
Terminal payload	Transported support reaching the terminal state.	Computational payload.
Curvature proxy	Detached Forman-style graph-curvature diagnostic.	Observer only.
Spectral observers	Detached Laplacian spectrum statistics.	Observer only.
World diversity	Distribution of utilization across world geometries.	Descriptive observer.